1. Human Nutrition

1.1. Diet

Balanced Diet: A diet that includes the appropriate amounts of carbohydrates, proteins, fats and oil, vitamin (C and D), mineral ions (calcium and iron), fibre and water needed to support the body's growth, energy needs, and overall health:

- Diet-related to **age/gender/lifestyle**:
  - **Children Below 12:** Require more calcium
  - **Teenagers:** Highest calorie intake
  - **Adults:** Balanced meal with fewer calories
  - **Pregnant Women:** more iron, calcium
  - **Males:** Generally require more energy

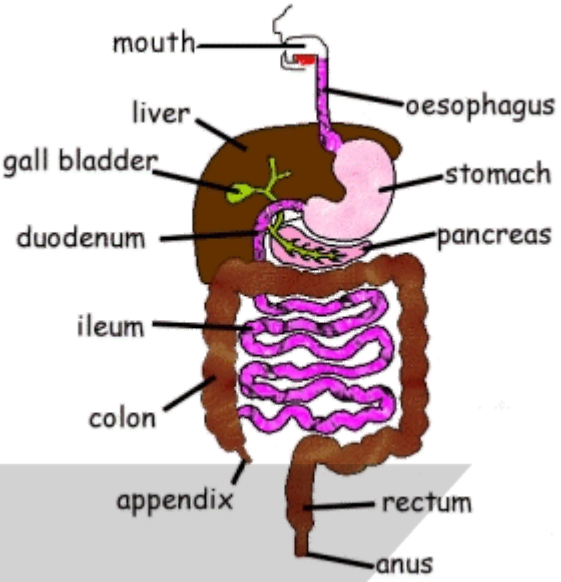
1.2. Nutrition

Nutrients	Uses	sources
Carbohydrates	Supply energy	Rice, bread, pasta, potatoes, maize, sugar, honey, fruits.
Fats and oils	Source of energy, building materials, energy store, insulation, buoyancy, making hormones	Butter, oil, nuts, seeds, avocados, fatty fish, meat
Proteins	Energy, building materials, enzymes, haemoglobin, structural material (muscle), hormones, antibodies	Meat, fish, eggs, beans, peas, lentils, milk, cheese, soy products, nuts
Vitamin C	Collagen, resistance to diseases	Citrus fruits, tomatoes
Vitamin D	Absorption of calcium	Fish liver oil, dairy, egg yolk, sunlight
Calcium	Development and maintenance of strong bones and teeth	Milk, cheese, leafy vegetables
Iron	Making haemoglobin	Red meat, liver, spinach, legumes
Fibre (Roughage)	Provides bulk for faeces, helps peristalsis	Vegetables, fruits, whole grains, bran
Water	Chemical reactions, solvent for transport	Drinks, fruits, vegetables, metabolic reactions

1.3. Deficiencies

- **Vitamin C:** *Scurvy;* loss of teeth, pale skin & sunken eyes
- **Calcium/Vitamin D:** *Rickets, Osteoporosis;* weak bones and teeth

1.4. Digestive System



Process of Digestion

- **Ingestion:** taking substances (e.g. food, drink) into the body through the mouth.
- **Physical/Mechanical Digestion:** breakdown of food into smaller pieces without chemical change.
  - It increases the **surface area of food** for the action of enzymes in chemical digestion.
- **Chemical Digestion**: breakdown of large, insoluble food molecules into small, soluble molecules.
- **Absorption**: the movement of nutrients from the intestines into the blood
- **Assimilation**: uptake and use of nutrients by cells
- **Egestion:** the removal of undigested food from the body as faeces

Main Organs in the Alimentary Canal

- **Mouth:** contains teeth used for mechanical Digestion, an area where food is mixed with salivary amylase & where ingestion takes place
- **Salivary glands:** produce saliva, which contains amylase and helps food slide down the oesophagus
- **Oesophagus:** tube-shaped organ that uses peristalsis (circular muscle contract and relax) to transport food from mouth to stomach
- **Stomach:** has **pepsin** (a protease that works best in **acidic conditions**) to break down proteins into amino acids and kills bacteria with hydrochloric acid. They also have elastic walls.
  - Mechanical digestion in the stomach **occurs via peristaltic contractions of the smooth muscle.**

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- **Small intestine:** tube-shaped organ composed of three parts:
  - **Duodenum:** fats are emulsified by bile and digested by pancreatic lipase to form fatty acids and glycerol. Pancreatic amylase and trypsin (a protease) break down starch.
  - **Jejunum** (not in syllabus)
  - **Ileum:** Maltase breaks down maltose to glucose. This is where absorption also takes place.
- **Pancreas:** produces amylase, trypsin (a protease that works best in **alkaline conditions**), and lipase.
- **Liver:** produces bile (emulsifies fats, neutralises acidic fat molecules), deamination, and makes urea to be sent to the kidney. Also, it is the site of the breakdown of alcohol and other toxins.
- **Gall bladder:** stores bile from the liver
- **Large intestine:** tube-shaped organ composed of two parts:
  - **Colon:** organ for absorption of minerals and vitamins and reabsorbing water from waste to maintain the body's water levels
  - **Rectum:** where faeces are temporarily stored
- **Anus:** a ring of muscle that controls when faeces is released.

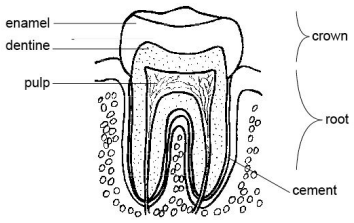
1.5. Teeth

Our teeth play a key role in **mechanical digestion** to help increase the surface area of food.

Incisors	Canines	Premolars	Molars
Rectangular shape, sharp for cutting and biting	Sharp-pointed for piercing and tearing	Blunt for chewing and grinding, one or two roots, cusps/bumps at the end	Blunt chewing and grinding. Two or three roots, ridges at the end

Structure of Tooth

Our teeth are embedded in bone, and the gums

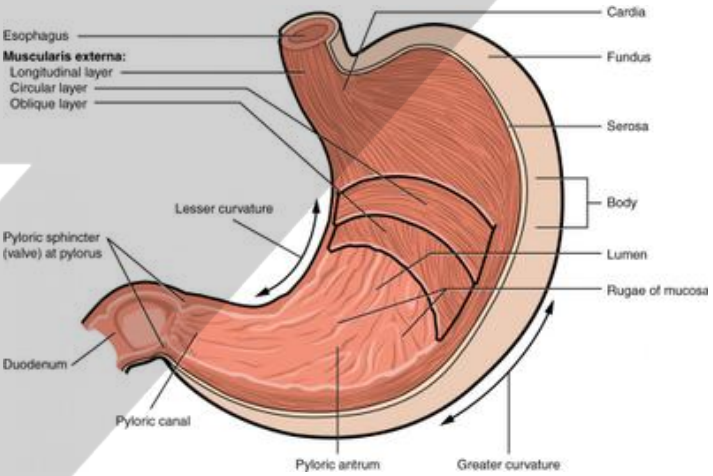


- **Enamel:** the strongest tissue in the body made from calcium salts
- **Cement:** helps to anchor tooth
- **Pulp:** contains tooth-producing cells, blood vessels, and nerve endings that detect pain.
- **Dentine:** calcium salts deposited on a framework of collagen fibers
- **Nerves:** Detect sensation
- **Blood vessels:** Provide nutrition

1.7. The Stomach

The stomach lining contains many smooth muscles that contract to squeeze physically and mix the food with the strong digestive juices present, a process also known as "stomach-churning."

- The food will be digested within the stomach for several hours
- First stop for **protein digestion**

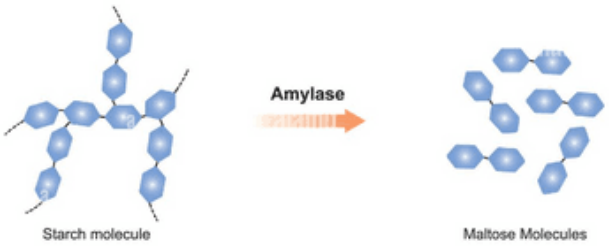


1.8. Chemical Digestion

- **Chemical Digestion**: the breakdown of large insoluble molecules into small soluble molecules
- The role of chemical digestion is to produce small soluble molecules that can be absorbed

1.9. Enzymes in Digestion

- **Amylase:** breaks down starch into maltose; it is produced in the pancreas (secreted into the duodenum) and salivary glands.



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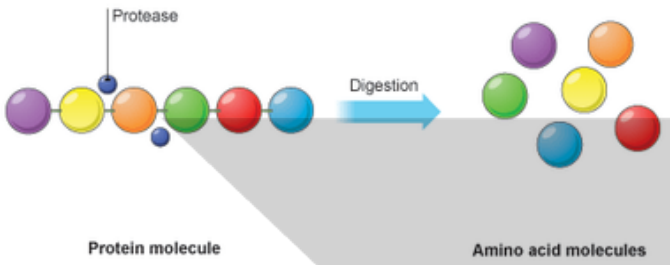
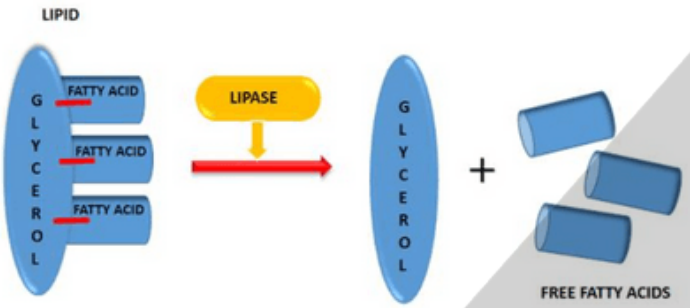
- **Maltase**: breaks down maltose into glucose in the membrane of the epithelium lining in small intestines.

- **Protease:** breaks down proteins into peptides and amino acids. Pepsin comes from the stomach (acidic), and trypsin comes from the pancreas (alkali).

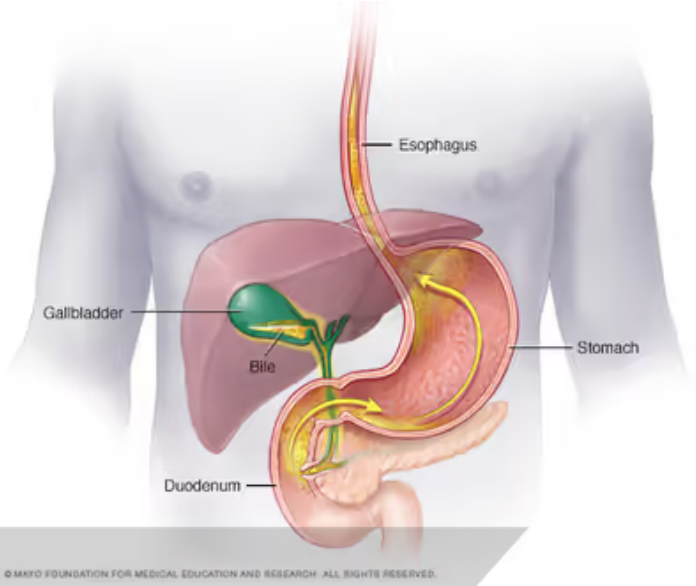


- **Lipase:** breaks down lipids into fatty acids and glycerol, produced by the pancreas.



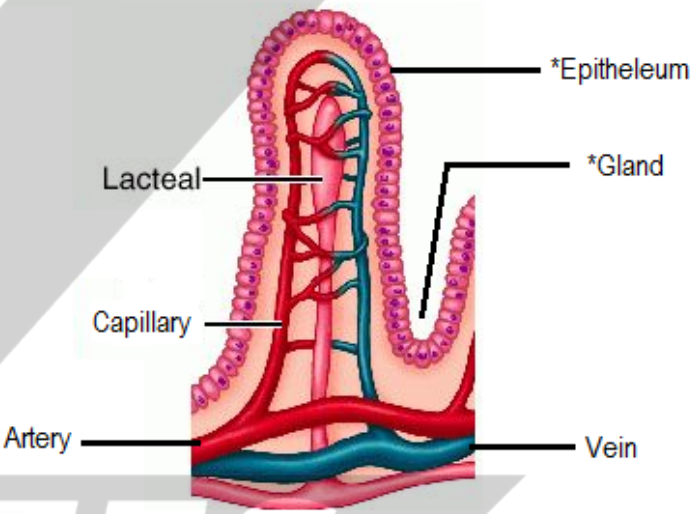
1.10. Hydrochloric Acid and Bile

- **Functions of Hydrochloric acid in gastric juice:**
  - Denaturing enzymes in harmful microorganisms.
  - Giving the optimum pH for pepsin activity
  - Kills pathogens
- **Bile**:
  - Produced by the liver
  - Stored in the gall bladder, transported by the bile duct
  - an alkaline mixture that neutralises the acid mixture of food and gastric juices entering the duodenum from the stomach to provide a suitable pH for enzyme action.
  - Bile helps with the digestion of fats by breaking them down from large molecules to smaller molecules, making a larger surface area.



1.11. Absorption and Villus

Absorption: the movement of nutrients from the intestines into the blood



- The small intestine is the region for **absorption of digested food** through diffusion, osmosis, and active transport.
- The small intestine is folded into many villi, increasing the **surface area for absorption**. One villus will have tiny folds on the cells on its outside called **microvilli**.
- The epithelium is **one cell thick**, allowing **efficient diffusion of nutrients**.
- A **large surface area** means more absorption of nutrients can happen.
- **Lacteals:** absorbs fatty acid and glycerol
- **Capillaries:** provide a good blood supply and a steep concentration gradient.
- **Goblet Cells:** Produces mucus
- Most water is absorbed from the small intestine, and some from the colon (large intestine).

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